

a) CALCULATION BY CONDUCTION (U-value)

Step 1: Calculate Basic Window U-value (U_h)

Formula from DB-HE:

$$U_h = (A_{\text{glass}} \times U_{\text{glass}} + A_{\text{frame}} \times U_{\text{frame}} + A_{\text{spacer}} \times \Psi_{\text{spacer}}) / A_{\text{total}}$$

Where:

- A_{glass} = Area of glass panes
- U_{glass} = U-value of glass (from manufacturer or Table 11)
- A_{frame} = Area of frame
- U_{frame} = U-value of frame (typically 1.4-2.8 W/m²K)
- A_{spacer} = Length of spacer around glass perimeter
- Ψ_{spacer} = Linear thermal transmittance of spacer (typically 0.04-0.08 W/mK)
- A_{total} = Total window area (A_{glass} + A_{frame})

Step 2: Apply Frame Correction Factor

From Table 11 note:

- Frame fraction F_F = 0.25 (simplified value)
- Adjusted U-value: U_{h,adjusted} = U_h × (1 + F_F)

Step 3: Verify Against Maximum Limits

Check against Table 3.1.1.a-HE1 limits:

Winter Climate Zone	Maximum U _h [W/m²K]
α	3.2
A	2.7
B	2.3
C	2.1

Winter Climate Zone	Maximum U_h [W/m ² K]
D	1.8
E	1.8

b) CALCULATION BY RADIATION (Solar Heat Gain)

This is more complex and involves multiple steps:

Step 1: Calculate Base Solar Transmittance ($g_{gl;wt}$)

Use Table 11 values:

Glass Type	$g_{gl;wt}$
Single glass	0.77
Double glass	0.68
Double low-e	0.60
Triple low-e	0.45

Step 2: Apply Movable Shading Correction

Use Table 12 based on:

- Shading type (exterior/interior)
- Color (white, pastel, dark, black)
- Glass type
- Shading properties (τ_0, B values)

Example: Double low-e glass with exterior white blinds:

$$g_{gl;sh;wt} = 0.03 \text{ (from Table 12)}$$

Step 3: Calculate Monthly Average with Shading Usage

Use Formula from Section 2.2.3:

$$g_{gl;wt;m} = (1 - f_{sh;wt;m}) \times g_{gl;wt} + f_{sh;wt;m} \times g_{gl;sh;wt}$$

Where:

- $f_{sh;wt;m}$ = Monthly activation fraction from **Tables 15.a/15.b**
- Based on: Climate zone, orientation, month

Step 4: Apply Fixed Obstruction Factors

Use Tables 16-19 based on obstruction type:

For Overhangs (Table 16):

Calculate ratios:

- L/H = Overhang projection / Window height
- D/H = Distance from window top to overhang / Window height

For Recesses (Table 17):

Calculate ratios:

- R/H = Recess depth / Window height
- R/W = Recess depth / Window width

For Louvers (Table 18):

Use values based on:

- Orientation
- Tilt angle (horizontal/vertical)

Step 5: Final Solar Transmittance Calculation

Complete Formula:

$$g_{total} = g_{gl;wt;m} \times F_{sh;obst} \times (1 - F_F)$$

Where:

- $g_{gl;wt;m}$ = Monthly average with movable shading
- $F_{sh;obst}$ = Obstruction factor from Tables 16-19
- F_F = Frame fraction (0.25 simplified)

PRACTICAL EXAMPLE

Let's calculate for a **south-facing window in Madrid (Zone D3)**:

Conduction (U-value):

Glass: Double low-e, $U_{glass} = 1.6 \text{ W/m}^2\text{K}$
 Frame: PVC, $U_{frame} = 1.8 \text{ W/m}^2\text{K}$
 Areas: $A_{glass} = 1.2 \text{ m}^2$, $A_{frame} = 0.3 \text{ m}^2$, $A_{total} = 1.5 \text{ m}^2$

$U_h = (1.2 \times 1.6 + 0.3 \times 1.8) / 1.5 = 1.64 \text{ W/m}^2\text{K}$
 Check: $1.64 < 1.8$ (limit for Zone D3) → COMPLIES

Radiation (July calculation):

Base glass: $g_{gl;wt} = 0.60$ (double low-e)
 With exterior white blinds: $g_{gl;sh;wt} = 0.03$ (Table 12)
 July activation: $f_{sh;wt;m} = 0.63$ (Table 15.a, Zone D3, South, July)

$g_{gl;wt;m} = (1-0.63) \times 0.60 + 0.63 \times 0.03 = 0.37 \times 0.60 + 0.63 \times 0.03 = 0.222 + 0.019 = 0.241$

With overhang (L/H=1, D/H=0.3): $F_{sh;obst} = 0.39$ (Table 16)
 Frame correction: $F_F = 0.25$

$g_{total} = 0.241 \times 0.39 \times (1-0.25) = 0.241 \times 0.39 \times 0.75 = 0.070$

Final Results:

- **Conductive U-value:** $1.64 \text{ W/m}^2\text{K}$
- **Radiative g-value (July):** 0.070

KEY DIFFERENCES SUMMARY

Aspect	Conduction Calculation	Radiation Calculation
Purpose	Heat loss/gain through temperature difference	Solar heat gain through radiation
Main Parameter	U-value [W/m²K]	g-value [dimensionless]
Time Dependency	Constant	Monthly variable
Shading Consideration	Not considered	Critical (movable + fixed)
Orientation Impact	None	Significant
Climate Zone Impact	Only for compliance limits	Direct calculation input

This methodology ensures compliance with Spanish DB-HE regulations while accurately representing the window's thermal performance for both energy loss and solar gain considerations.